

Towards a FAIR Framework for Earthquake catalogues in Aotearoa New Zealand

A Community White Paper

Making New Zealand's earthquake catalogues Findable, Accessible, Interoperable, and Reusable

Version 0.1 — Draft for review

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Executive Summary

Earthquake catalogues underpin seismology, monitoring, engineering, and risk management. Aotearoa New Zealand produces many of them — the GeoNet national catalogue alongside regional, volcano-specific, temporary-deployment, historical, and derived catalogues — but they are fragmented: built to different standards, documented inconsistently, and rarely interoperable. The 2022 National Seismic Hazard Model had to re-standardise magnitudes across sources before it could use them (Gerstenberger et al., 2024); high-value research catalogues such as the 86,579-event Taupō Volcanic Zone catalogue (Ilsley-Kemp and Mestel, 2025) sit outside any common system; and location quality across the national catalogue varies markedly with station coverage (Warren-Smith et al., 2025). Combining catalogues today is a manual, error-prone task.

Two national workshops (2023 and 2024) brought together researchers, agencies, and industry to co-design a practical framework guided by the FAIR principles (Findable, Accessible, Interoperable, Reusable) and, in the Aotearoa New Zealand context, by the CARE Principles for Indigenous Data Governance (Collective benefit, Authority to control, Responsibility, and Ethics). This white paper sets out the current landscape, the community-identified gaps, and a proposed national framework: a “catalogue of catalogues” that *complements rather than replaces* GeoNet. The framework pairs a lightweight registry of catalogue descriptors with a platform of FAIR services built on it, and is accompanied by governance, a roadmap, and an indicative resourcing envelope. A working prototype demonstrates the end-to-end workflows (ingestion, quality assessment, merge and conflation, analytics, and programmatic access). We ask New Zealand’s earth-science community, its agencies, and its research-infrastructure funders to endorse the standards proposed here and to help establish a sustained national capability.

Key Recommendations

1. Adopt the agreed minimum-attributes *submission profile* and a controlled parameter lexicon, aligned with the international QuakeML and FDSN seismic-data standards, so every catalogue is described consistently (distinguishing required from recommended fields).
2. Stand up the lightweight *Registry* first — the openly versioned “catalogue of catalogues” of catalogue descriptors — as the low-barrier, near-term community deliverable for discovery.
3. Build the *Platform* of FAIR services incrementally on top — programmatic API, cross-agency event-identifier mapping, transparent merge/conflation, configurable quality scoring, and analytics — for catalogues ingested in full, complementing rather than replacing GeoNet.
4. Require persistent identifiers (DOIs), explicit versioning, and machine-readable changelogs, with universal uncertainty reporting (and qualitative confidence classes for historical events), so any cited catalogue version is reproducible.
5. Establish formal governance — a Steering Committee (with substantive Māori representation) and a Data Stewardship Team — that operationalises both FAIR and CARE principles, underpinned by sustained multi-year funding.

Keywords: earthquake catalogues; FAIR principles; CARE principles; seismology; research data infrastructure; data governance; Aotearoa New Zealand.

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1 Introduction: The Problem and the Opportunity

Earthquake catalogues are the cornerstone of seismology and the shared evidence base for hazard models, engineering design, rapid response, and fundamental research (Woessner et al., 2010). Their value depends on being trustworthy, comparable, and reusable.

In Aotearoa New Zealand that evidence base is fragmented. GeoNet curates the authoritative national catalogue, with broad coverage and open access through web and programmatic services (GeoNet, 2025a,c,d); but its location quality and completeness vary markedly across the country with station density (Warren-Smith et al., 2025), and metadata conventions have evolved over time. Alongside it sit many high-value catalogues that are effectively invisible to the national system:

- regional and volcano-specific catalogues — for example, the consistent, high-precision Taupō Volcanic Zone catalogue of 86,579 events (Illsley-Kemp and Mestel, 2025);
- temporary aftershock and ocean-bottom deployments with lower magnitude thresholds;
- historical and pre-instrumental compilations from written and oral records; and
- derived and synthetic catalogues built for hazard and risk modelling.

Because these catalogues use different formats, identifiers, magnitude scales, and metadata, combining them is manual and error-prone. The 2022 National Seismic Hazard Model illustrates the cost: a flagship national product required a dedicated effort to standardise magnitudes across sources before they could be used (Gerstenberger et al., 2024). The consequences of the status quo are lost researcher time, reduced reproducibility, and avoidable uncertainty in decisions that bear on life safety and billions of dollars of infrastructure.

The opportunity is to add a thin coordinating layer — a “catalogue of catalogues” — that makes these datasets findable, comparable, and citable without replacing or duplicating GeoNet’s operational mandate. In practice this layer has two parts: a lightweight *registry* of catalogue descriptors (for findability) and a *platform* of richer FAIR services — comparison, merging, and citable versions — built on top of it. Internationally, initiatives such as the European Plate Observing System (EPOS) Seismology service show the value of standardised, FAIR data services and coordinated infrastructure (EPOS, 2025b,a). This paper synthesises outcomes from two national workshops and intersession working-group activity to propose such a framework, tailored to New Zealand and governed under both FAIR and CARE principles.

This paper has three aims: (i) to characterise the landscape of New Zealand’s earthquake catalogues and benchmark it against mature international systems; (ii) to propose a common set of minimum attributes for describing and creating catalogues; and (iii) to set out a gap analysis and near-term community steps toward a federated, harmonised national catalogue capability for earthquake science and engineering.

The framework is shaped by concrete user needs identified at the workshops: gaining an overview of which catalogues exist and where to find them; locating catalogues for a specific region; merging and comparing catalogues; and identifying which catalogues meet the requirements of a given analysis (for example, those that include phase picks, not just hypocentres). Figure 1 contrasts the current and proposed states.

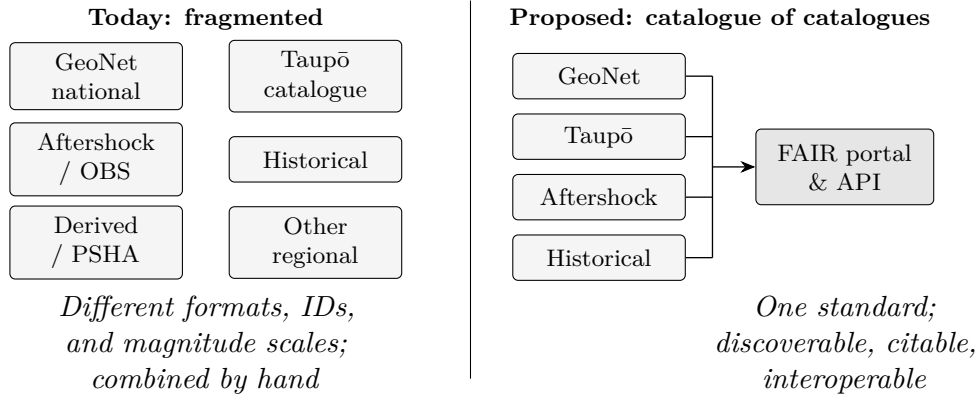


Figure 1: The problem and the opportunity. Today, New Zealand’s earthquake catalogues are developed independently and combined manually. The proposed framework adds a coordinating layer that makes them interoperable through shared standards, a common portal, and a programmatic API — complementing GeoNet rather than replacing it.

2 The Current Landscape and International Benchmark

New Zealand’s earthquake data ecosystem comprises several catalogue types that serve different purposes. The **national operational catalogue**, maintained through GeoNet by Earth Sciences New Zealand (ESNZ; formerly GNS Science), provides consistent coverage and open access since the 1930s (GeoNet, 2025a,c; GNS Science, 2020); its gaps are evolving metadata conventions, mixed magnitude types, spatially variable location quality (Warren-Smith et al., 2025), and limited exposure of detailed processing fields. **Regional and volcano-specific catalogues** add high-resolution detail using dense networks or specialised methods (Illsley-Kemp and Mestel, 2025) but are often not interoperable with national standards. **Temporary deployment** catalogues are valuable yet not routinely discoverable. **Historical, derived, and synthetic** catalogues complement the instrumental era but, without formal versioning, identifiers, and processing documentation, are difficult to reuse or compare (Gerstenberger et al., 2024). In short, the ecosystem’s strengths lie in local precision and national consistency, while its gaps lie in completeness through time and space, processing consistency, magnitude and depth scales, metadata, transparency, and access to legacy and temporary data.

This ecosystem is large and heterogeneous. Table 1 illustrates the main catalogue types contributed by the community, with representative examples and their characteristic strengths and limitations.

Table 1: Representative earthquake-catalogue types in Aotearoa New Zealand, with examples and characteristic strengths and limitations (from the community workshops).

Type	Examples	Strengths / limitations
Volcano-seismic / microseismic	Taupō, Whakaari, Ruapehu, geothermal fields	Precise depths, uniform local methods; limited spatial scope, weak interoperability
National operational	GeoNet rapid and reviewed, GeoNet centroid moment tensors (CMT), strong-motion datasets	Consistency, continuity, accessibility, manual review; evolving conventions, mixed magnitude types
Local / regional deployments	Alpine Fault OBS, Hikurangi (e.g. HOBITSS), Kaikōura/Cook Strait aftershocks, TVZ experiments (SAMBA, COSA, PTANGO)	High-quality velocity models, precise locations and depths; short duration, limited discoverability
Source-mechanism	Ristau moment tensors, Global Centroid Moment Tensor (GCMT), finite-fault and stress-drop inversions	Kinematic and stress information for hazard and tectonics; standalone, weakly integrated
Historical / pre-instrumental	Pre-1940 compilations, oral and written records, NSHM inputs	Extend the record by centuries; large uncertainties, poor depth/location, few small events
Slow-slip, palaeo-seismic, geodetic	Slow-slip detections, palaeo-seismic records, offshore geodetic detections	Unique long-timescale and aseismic information; sparse coverage, imprecise timing
Research / method-specific	Template-matching, machine-learning, relocation, and waveform-inversion catalogues	Gains in completeness and precision, documented methods; bespoke formats, variable access
Derived / international	NSHM-derived snapshots; ISC, ComCat, EMSC combined catalogues	Broad or standardised coverage; not NZ-specific, variable completeness

Mature international systems show what “good” looks like: the USGS Advanced National Seismic System Comprehensive Earthquake Catalog (ComCat) federates 100+ networks (USGS, 2025); the Swiss Seismological Service couples a dense network with cross-border data (Diehl et al., 2025); the European-Mediterranean Seismological Centre (EMSC) and the International Seismological Centre (ISC) aggregate across many agencies (European-Mediterranean Seismological Centre, 2025b; International Seismological Centre, 2025a); and EPOS provides standardised FAIR services across Europe (EPOS, 2025b). These systems share common standards — notably the International Federation of Digital Seismograph Networks (FDSN) web services and the QuakeML data format — which New Zealand already supports. Appendix A summarises each. Table 2 benchmarks New Zealand against these systems across ten dimensions and identifies the priority gaps: (1) universal uncertainty reporting; (2) formal version control with Digital Object Identifiers (DOIs) and changelogs; (3) a cross-agency event-identifier mapping service; (4) integration of historical and pre-instrumental data; and (5) formal governance with dedicated funding. These gaps motivate the framework components and Key Recommendations that follow, though not in strict one-to-one correspondence.

Table 2: Comparison of NZ and International Earthquake Catalogue Capabilities

Feature	New Zealand	USGS ComCat	Swiss SED / EPOS	Gaps & Priorities
Core parameters	Origin time, location, depth, magnitude, phase picks, station counts	Hypocenter, magnitudes, phase data, moment tensors, macroseismic data (USGS, 2025)	Historical + instrumental events, uniform M_w , uncertainties, intensities (Diehl et al., 2025)	Lacks full uncertainty propagation for all events
Metadata & provenance	Some document processing pipelines; incomplete or inconsistent across catalogues	Maintains standards, data dictionaries, method documentation (USGS, 2025)	Adheres to FDSN, StationXML, QuakeML standards (Diehl et al., 2025)	Need consistent, centralised metadata policies
Versioning & review	Rapid + review workflows; some reprocessing	Versioned updates, multiple origins per event, revisions over time (USGS, 2025)	Unified access to evolving datasets, harmonisation	Formalise version control (DOIs, changelogs)
Access & APIs	Quake Search, FDSN, QuakeML, CSV/GeoJSON export	FDSN API, GeoJSON/CSV feeds, search interfaces (U.S. Geological Survey, 2025)	Integrated access to waveforms, parametric data via standards (EPOS, 2025b)	Enhance federated discovery, cross-catalogue queries
Historical data	Some pre-instrumental events with greater uncertainty	Includes large historic events, variable completeness	Centuries of data, macroseismic fields, uncertainties (Diehl et al., 2025)	Strengthen integration, metadata, uncertainty modelling

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Table 2 – *continued*

Feature	New Zealand	USGS ComCat	Swiss SED / EPOS	Gaps & Priorities
Uncertainty & quality	Many provide estimates (location/magnitude error); not universal	Quality metrics: RMS, station count, azimuthal gap (USGS, 2025)	Emphasises uncertainty, solution quality, EPOS standards (Diehl et al., 2025)	Universal uncertainty reporting needed
Multi-catalogue linking	Mostly separate; limited cross-catalogue mapping	Aggregates many networks, links multiple origins per event (USGS, 2025)	Harmonised catalogues across Europe; cross-border exchange (EPOS, 2025b)	Build Event ID mapping service, federated portal
Special products	Focal mechanisms, moment tensors, slow slip as standalone datasets	Hosts moment tensors, additional products tied to events (USGS, 2025)	Integrates fault, slip, seismological products	Better integrate special-purpose catalogues
Standards compliance	Supports FDSN/QuakeML; not all catalogues fully adhere	FDSN and QuakeML in APIs and exports (U.S. Geological Survey, 2025)	Full QuakeML, StationXML, FDSN compliance (Diehl et al., 2025)	Ensure full compliance across all catalogues
Governance & sustainability	ESNZ/GeoNet or individual researchers; less formal structure	Formal governance, funding, institutional roles (USGS/ANSS/NEIC) (USGS, 2025)	Robust institutional backing, cross-national coordination	Establish formal governance, roles, funding

3 A Community-Driven Process

The framework was co-designed with the community. Two pre-conference workshops at the Geoscience Society of New Zealand annual meetings provided the primary engagement: a 2023 workshop framed needs and objectives, and a 2024 workshop reviewed progress and refined priorities (Geoscience Society of New Zealand, 2023, 2024). Participants spanned university researchers, GeoNet and Earth Sciences New Zealand staff, government agencies, and engineering and insurance users. Discussion was structured around eight discussion tables — covering minimum catalogue attributes, metadata and documentation, access and versioning, governance, and community priorities — whose outcomes are summarised in Appendix C. Between workshops, a working group met regularly to draft standards, test integration pathways, and build a prototype portal.

3.1 Prototype Portal: Proof of Concept

To ground the framework in a working system, the working group built a functional prototype portal — *Catalogue of Catalogues* (CofC) — and used it at the 2024 workshop to demonstrate the Platform-layer services (Section 4) — ingestion, merge, export, and analytics — with representative New Zealand datasets. The prototype is open-source (MIT licence) and archived with a citable release (see Data and Code Availability); we summarise it here as evidence of feasibility rather than as a validated production service. The capabilities below are implemented in the prototype, whereas the national framework in Section 4 is a proposal that, except where noted, is not yet implemented. The prototype demonstrates:

- **Ingestion** of CSV/tab-delimited text (with automatic delimiter and date detection), JSON/GeoJSON variants, and QuakeML 1.2, with an alias-based field-mapping interface and direct, de-duplicating sync from GeoNet’s FDSN Event service;
- **Search and visualisation** — spatial (bounding box, polygon, radius), time, magnitude, and depth filtering on an interactive map with event clustering, uncertainty ellipses, focal mechanisms, and station coverage;
- **Merge and conflation** with five configurable strategies, conflict detection and logging, lineage tracking, and a pre-commit preview (Section 4.4);
- **Analytics** — Gutenberg-Richter b -value (Aki, 1965), completeness magnitude (M_c) (Wiemer and Wyss, 2000), cumulative seismic-moment release, Gardner-Knopoff declustering (Gardner and Knopoff, 1974), and multi-catalogue magnitude-frequency comparison; and
- **Multi-format export** (CSV, GeoJSON, KML, QuakeML 1.2), including version-specific exports.

Community feedback from the 2024 workshop informed refinements to the metadata schema, merge options, and quality-grading approach described next.

4 The Proposed FAIR Framework

The framework addresses the end-to-end lifecycle of catalogue data; Figure 2 summarises how its components fit together. This section describes the framework conceptually. Implementation-level detail — API endpoints, access-control roles, field-level representations, and validation thresholds — is collected in Appendix B so that this section stays readable for a general audience.

The framework is organised as two complementary layers. The **Registry** — the “catalogue of catalogues” in the literal sense — is a lightweight, openly versioned register of catalogue *descriptors*: each catalogue is summarised by a short submission profile (Section 4.2) recording what it contains and how to obtain it, without transferring the underlying data. It is cheap to contribute to and is the immediate community deliverable, answering *which catalogues exist, what they contain, and where to find them*. The **Platform** provides the richer FAIR services — programmatic access, cross-catalogue event-identifier mapping, merge and conflation, quality scoring, analytics, and versioned DOIs — for catalogues that are ingested in full, answering *compare, merge, analyse, and cite*. The submission profile is the common front door: a catalogue can remain a Registry descriptor that links to wherever it lives, or opt in to ingestion for the full Platform services. The Registry is available first and at low cost; the Platform is demonstrated by the prototype (Section 3.1) and built out across the roadmap (Section 5). The components below describe both layers.

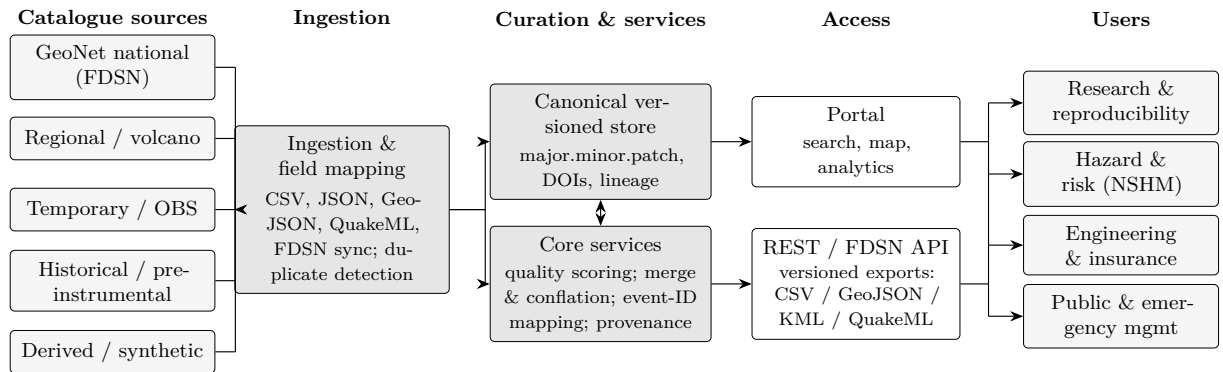


Figure 2: End-to-end architecture of the proposed framework. Heterogeneous catalogue sources are ingested and mapped to a canonical, versioned store; curation services (quality scoring, merge and conflation, event-identifier mapping, and provenance) feed a portal and a programmatic API that serve research, hazard, engineering, and public users. Capabilities identified as implemented in Section 3.1 are exercised end-to-end by the prototype. The figure details the *Platform* layer; the *Registry* — the lightweight layer of catalogue submission profiles (Table 3) through which catalogues are first described and discovered — is the low-cost, Registry-first entry point (Section 4) and feeds optional full ingestion into the Platform shown here.

4.1 Metadata Standards and a Shared Lexicon

A core metadata profile aligned with QuakeML and FDSN conventions captures identifiers, origin parameters with uncertainties, magnitude types with errors, detection and quality metrics, methodology descriptors, event classification, and catalogue-level metadata (ETH Zürich Seismology / QuakeML Working Group, 2015; GeoNet, 2025c; U.S. Geological Survey, 2025). Required versus recommended fields are distinguished to balance utility against effort; the full field hierarchy is given in Table 6 (Appendix B). A controlled vocabulary standardises magnitude types, event classifications, phase names and quality codes, units and formats, and dataset naming, improving interpretability and filtering across catalogues (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016). An illustrative subset of this lexicon is given in Table 7 (Appendix B).

4.2 Catalogue Submission Profile (Minimum Attributes)

Every catalogue in the Registry is described by a concise *submission profile* — the minimum set of attributes agreed at the 2024 community workshop. The profile records, consistently and

searchably, *what* a catalogue contains and *how* to obtain it, so that users can discover catalogues and judge fitness for purpose without first reading each catalogue’s documentation. Required fields are kept to a small core to lower the barrier to contribution, and controlled drop-downs are preferred over free text wherever possible to keep entries searchable and consistent. The profile complements the event-level metadata profile (Table 6, Appendix B): the profile describes a catalogue *as a whole*, whereas the metadata profile describes the recommended internal structure of a good catalogue. Table 3 lists the fields.

Table 3: Catalogue submission profile: minimum attributes for a catalogue-of-catalogues entry, as agreed at the 2024 workshop. Controlled options are shown in this style; fuller detail is carried by the linked dataset/DOI.

Field	Status	Options / notes
<i>Identification and contact</i>		
Catalogue name	Required	Short, human-readable title
Contact person(s)	Required	At least one; name and email, multiple separated by semicolons
Data source	Recommended	Origin of the catalogue (e.g. parent catalogue, instruments, records)
Brief description	Required	Why the catalogue was produced and its focus; placed at the end of the form, with a character limit
Additional notes	Optional	Free-text notes/keywords; placed last
<i>Coverage</i>		
Start date	Required	
End date	Optional	Or tick continuous/ongoing
Bounding box	Required	Latitude/longitude (N/S/E/W)
Depth range	Recommended	Minimum and maximum depth
Region of interest	Recommended	Descriptive focus or extent
Magnitude info	Optional	Magnitude range and/or magnitude of completeness — not required (many historical/intensity catalogues lack instrumental magnitudes)
<i>Content present</i> (tick all that apply)		
Content flags	Recommended	hypocentral locations; relative relocations; phase arrival-time picks; amplitude picks; fault-geometry (non-point-source); historical observations (felt reports / macroseismic intensities); quality information; other
<i>Classification</i> (controlled drop-downs)		
Catalogue type	Recommended	background / aftershock / induced / historical / compilation / large-events-only
Focus	Recommended	geographical / parameter-testing / particular-event-type
Detection method	Recommended	manual / automatic / template-matching / machine-learning / felt-reports / other
Review status	Recommended	non-reviewed / partially reviewed / reviewed
<i>Availability</i>		
Data availability	Required	available (DOI) / available on request / not available

Consistent with this scope, the Registry entry stays high-level. Deeper detail is carried by the catalogue’s own dataset or publication and referenced by DOI, rather than duplicated in the Registry. This includes velocity models, station and pick information, processing provenance and parameters, waveform and raw-data links, focal-mechanism details, input-catalogue lineage, links to GeoNet event IDs, and any associated-publication abstract. A graphical, map-based discovery interface (polygon selection and faceted filtering) and contributor self-service editing are planned as future enhancements; in the interim, Registry entries are openly searchable and exportable, while catalogues ingested into the Platform additionally gain the services described below.

4.3 Interoperability: Programmatic Access and Cross-Catalogue Identifiers

A unified API provides machine-readable access across catalogues, returning CSV, GeoJSON, and QuakeML, with bulk downloads of versioned datasets to support reproducible research; it builds on existing GeoNet interfaces and standards (GeoNet, 2025d,c; ETH Zürich Seismology / QuakeML Working Group, 2015). The endpoint design is given in Appendix B.

A persistent interoperability challenge is that the same earthquake carries different public identifiers across agencies — a New Zealand event may appear as a GeoNet Public ID, a USGS event code, and an ISC event number — so researchers aggregating sources must de-duplicate by hand. The framework includes an event-identifier mapping service that maintains cross-agency equivalences, matched using configurable spatial, temporal, and magnitude thresholds (illustratively, separation < 20 km, time offset < 30 s, magnitude difference < 0.5). For comparison, the ANSS composite catalogue identifies duplicates on space and time alone, using operational windows of 16 s and 100 km (Northern California Earthquake Data Center, 2024); the magnitude criterion is added here to reduce false matches in dense sequences. These values are deliberately permissive to absorb differing location procedures; because wide windows can group genuinely distinct events in dense aftershock sequences, thresholds are configurable per source and ambiguous matches are flagged for expert review rather than merged automatically. The service aligns with the approach used by EMSC (European-Mediterranean Seismological Centre, 2025b) and extends it to the New Zealand context.

4.4 Catalogue Merge and Conflation

Catalogue conflation — combining events from multiple independent catalogues into a unified view — is a central operation that builds on established cross-agency practice (Si et al., 2024; Northern California Earthquake Data Center, 2024). The framework specifies five merge strategies, all implemented in the prototype (Section 3.1); users choose by scientific objective:

- **Quality-based** (default): the highest quality-scored solution (Section 4.5) is retained.
- **Priority-based**: a nominated authoritative agency’s solution is preferred; others are recorded as alternatives.
- **Newest data**: the most recently determined solution is retained, where later analyses supersede earlier ones.
- **Most complete**: the solution with the most populated fields is retained, maximising metadata density.
- **Averaged**: location, depth, and magnitude are combined across solutions by inverse-variance weighting (with an ISC magnitude-type hierarchy to avoid scale saturation). Because agency solutions for one event share data and processing assumptions and are therefore not statistically independent, the merged uncertainty is reported conservatively — not

below the smallest contributing value — rather than the smaller independent-combination result that would follow from assuming the solutions are statistically independent.

Conflicts are detected and logged when solutions disagree beyond configurable thresholds (spatial separation, temporal offset, magnitude difference, depth range, network mismatch); duplicate detection during dense aftershock sequences follows nearest-neighbour methods (Vorobieva et al., 2022). Each merged event retains explicit lineage to its source events and the configuration used, enabling full audit and reproducibility; a dry-run preview returns conflict statistics without committing a new version. Table 4 shows a worked example.

Table 4: Worked example: averaged merge of one event reported by two agencies. Merged point values are inverse-variance-weighted means; the merged uncertainty is reported conservatively (see Section 4.4). Identifiers are real (the 2016 Kaikōura mainshock) but the tabulated parameters are simplified for illustration and are not the official published solutions.

Field	Source (GeoNet)	A	Source B (USGS)	Merged (uncertainty- weighted)
Event identifier	2016p858000		us1000778i	merged-0001 (lineage to A, B)
Origin time (s past 11:02:00 UTC)	56.3 ± 0.5		56.7 ± 0.8	56.4 ± 0.5
Latitude ($^{\circ}$)	-42.737		-42.740	-42.738
Longitude ($^{\circ}$)	173.054		173.100	173.063
Depth (km)	15.1 ± 2.0		15.0 ± 4.0	15.1 ± 2.0
Magnitude (M_w)	7.8 ± 0.1		7.8 ± 0.1	7.8 ± 0.1
Horizontal unc. (km)	1.5		3.0	1.5

4.5 Data Quality Assessment

A standardised, transparent quality score supports informed reuse decisions, building on established practice for assessing catalogue quality (Woessner and Wiemer, 2005) and on the criteria used to benchmark GeoNet location quality (Warren-Smith et al., 2025). Each event receives a score (0–100) computed as a weighted sum over five dimensions of solution quality:

- **Location** (35 %) — horizontal, depth, and time uncertainties, weighted most heavily on horizontal uncertainty;
- **Network geometry** (25 %) — azimuthal gap and used-station and used-phase counts, the primary instrumental predictors of location error (Bondár et al., 2004; Warren-Smith et al., 2025);
- **Solution fit** (15 %) — the RMS arrival-time residual of the location;
- **Magnitude** (15 %) — magnitude uncertainty and the number of contributing stations; and
- **Evaluation** (10 %) — manually reviewed/final solutions score above automatic/preliminary ones.

Each dimension penalises missing fields, so sparsely documented events are scored conservatively. Field completeness, physical-consistency, and magnitude-homogeneity checks are handled separately by the automated validation suite (Appendix B). The score maps to the grade bands in

Table 5; the weights and bands are configurable defaults rather than an empirically calibrated standard, and should be tuned by the community against representative New Zealand catalogues. Flagged statistical outliers are annotated for caution rather than removed automatically.

Table 5: Quality Grade Bands and Recommended Use (illustrative; subject to community calibration; full scheme in Appendix B)

Grade	Score	Interpretation and recommended use
A	85–100	Suitable for hazard modelling and quantitative analysis.
B	65–84	Acceptable; document limitations or supplement meta-data.
C	45–64	Poor; expert review recommended before quantitative use.
D/F	<45	Minimal utility; discovery or illustrative purposes only.

4.6 Versioning, Persistent Identifiers, and Provenance

Every catalogue release carries a version, timestamp, and changelog, and all versions remain accessible for reproducibility. The framework uses a three-level MAJOR.MINOR.PATCH scheme (worked examples in Table 8, Appendix B): *major* increments alter event parameters, *minor* increments add events or optional fields, and *patch* increments change only metadata or documentation.

Persistent identifiers make those releases citable. The catalogue series carries a single concept DOI, and each major and minor release — the changes that affect citable content — receives its own version DOI (DataCite, 2025); patch releases update the existing record without a new DOI, and frequent minor releases may be batched (for example, quarterly) to avoid identifier proliferation. A machine-readable changelog accompanies every release, so a researcher can cite the exact data state used in a study and a reviewer can see what changed between versions.

Processing steps (detection, picking, location, velocity models, magnitude calibration) and provenance are documented following the W3C PROV model of entities, activities, and agents, with software version and execution date as minimum requirements (World Wide Web Consortium, 2013). Uncertainties for time, location, and magnitude are reported as standard deviations (with confidence-ellipse parameters where anisotropic) and always carry the magnitude type code; events without quantified uncertainties are flagged, and historical events may carry a qualitative confidence class (e.g. “well-constrained”, “felt-only”). Field-level representations are specified in Appendix B.

4.7 Special Catalogue Types

Different catalogue types need adapted handling to preserve scientific value:

- **Regional research catalogues** are ingested as distinct, separately versioned catalogues and cross-referenced — not silently merged — with the national catalogue, because different processing assumptions can introduce systematic biases.
- **Temporary deployments** are labelled with their time window and station geometry.
- **Historical and pre-instrumental catalogues** use qualitative confidence classes and macroseismic intensity fields (Modified Mercalli or EMS-98, Grünthal, 1998), and form a distinct collection.
- **Derived and synthetic catalogues** carry explicit provenance to their source versions.

Appendix B gives the detailed handling.

4.8 Licensing and Access; FAIR Compliance Indicators

To support FAIR reusability, every dataset and metadata record should carry an explicit, machine-readable licence (Wilkinson et al., 2016; GO FAIR Initiative, 2025). For government-funded baseline products, a default open licence is recommended, aligned with New Zealand Government Open Access and Licensing (NZGOAL) guidance (New Zealand Government, 2022), with documented exceptions for embargoed, sensitive, or culturally restricted data. Access is governed by a role-based model with catalogue-level public/registered/restricted designations and, where needed, field-level masking of culturally sensitive locations; all write operations are audited. These mechanisms are detailed in Appendix B. Progress should be tracked against a FAIR compliance matrix that maps each FAIR sub-principle (F1–F4, A1–A2, I1–I3, R1.1–R1.3) to a framework mechanism and a measurable, annually reported indicator — for example, the percentage of catalogues with a resolvable DOI (F1), mean metadata-completeness (F2), API uptime (A1), and the percentage carrying a machine-readable licence (R1.1). The full matrix is given in Table 11 (Appendix B).

4.9 Aotearoa New Zealand Data Governance: FAIR with CARE

In Aotearoa New Zealand, openness operates within Te Tiriti o Waitangi: the Crown–Māori partnership creates obligations of governance, protection, and active participation that extend to data about, or of significance to, tangata whenua. Earth Sciences New Zealand is a Crown research entity, so the framework must give effect to those obligations, applying the CARE Principles for Indigenous Data Governance (Carroll et al., 2020) alongside FAIR and consistent with Māori data-sovereignty practice (Kukutai and Taylor, 2016). Concretely, the four CARE principles are operationalised as follows. *Collective benefit*: engagement and benefit-sharing arrangements for catalogues of significance to iwi and hapū. *Authority to control*: iwi and hapū determine the access and use conditions for their data — the portal’s access controls (Appendix B) only *implement* conditions that iwi and hapū set, and never substitute for that authority. *Responsibility*: documented engagement pathways, attribution, and reporting back to communities. *Ethics*: review of culturally sensitive records before release. Here “mātauranga-informed” means catalogue records or interpretations that draw on Māori knowledge — for example, oral-tradition accounts of historical earthquakes or place-based knowledge of shaking and ground effects — which carry their own provenance and access expectations. For this commitment to be substantive rather than aspirational, iwi and hapū must be engaged as partners and decision-makers from the outset, and the Steering Committee (Section 5.1.1) must include Māori membership with a substantive governance role, not a single advisory seat.

5 Implementation: Governance, Roadmap, and Funding

5.1 Governance, Roles, and Responsibilities

5.1.1 Steering Committee

A Steering Committee provides strategic oversight and ensures community ownership. Membership should include Earth Sciences New Zealand, at least two universities with active seismology programmes, a government user (e.g. the National Emergency Management Agency), an engineering or insurance representative, Māori representation with the mandate to speak to

data-sovereignty matters (see Section 4.9), and, where feasible, an international advisor from a peer organisation (e.g. ISC, EMSC, or the GFZ Helmholtz Centre for Geosciences). It meets at least quarterly with defined decision-making processes and staggered three-year terms. The Committee approves new data sources for ingestion, ratifies versioning decisions affecting the national record, oversees international partnerships (e.g. alignment with EPOS), and commissions annual reporting against FAIR compliance indicators.

5.1.2 Data Stewardship Team

A Data Stewardship Team led by Earth Sciences New Zealand handles day-to-day operations: a Data Stewardship Lead, Data Managers (ingestion, metadata curation, quality review), a Software Engineer (API, infrastructure, security), and a User Support Analyst. External researchers may also submit catalogues through a defined contribution process (Section 5.1.3). These governance roles (contributor, reviewer, maintainer) are distinct from the portal’s technical access roles defined in Appendix B.

5.1.3 Catalogue Contribution Process

External researchers and institutions may submit catalogues for inclusion. Submissions require: catalogue name and description, responsible institution and contact, time period and geographic coverage, event count and magnitude range, format and metadata documentation, and a declared licence. A two-stage review process applies: technical QC by the Data Stewardship Team (format validity, completeness score, duplicate check) followed by scientific peer review by a standing panel of subject-matter seismologists drawn from the contributing institutions (methods assessment, comparison with existing catalogues). Because the Committee meets only quarterly, responsibility is divided to keep the contribution turnaround within a target of four to eight weeks: the Steering Committee sets ingestion policy and approves *categories* of source, while the Team approves individual contributions against that policy. Only novel or contentious cases are escalated to the Committee. Written feedback is provided regardless of outcome, and approved catalogues are assigned a DOI and version record.

5.1.4 Registry Custody, Updates, and Use Conditions

Each registered catalogue names a custodian — the person responsible for its entry — with a listed contact for updates, corrections, or withdrawal. The custodian need not be the original author, so that past studies and legacy datasets can be registered by others; in such cases the entry clearly attributes the catalogue’s authorship. Where the same catalogue might be registered more than once (for example, a legacy dataset entered independently), duplicate entries are identified and reconciled by the Data Stewardship Team. The Registry itself is version-controlled, with its release history maintained in a public Git repository, so that additions, edits, and deprecations are transparent and a superseding catalogue version can be linked to the entry it replaces.

The Registry is a finding aid, not an endorsement (the Registry submission profile itself is defined in Section 4.2). A clear disclaimer states that entries are contributed and not independently verified, and that the Registry simply records which catalogues exist. Users are encouraged to reference catalogues appropriately and to contact and collaborate with the original authors where relevant, consistent with the intellectual-property and data-sharing expectations attached to each dataset.

5.1.5 Success Metrics and Annual Reporting

Progress should be measured annually against indicators including: number of catalogues ingested; total event count and geographic coverage; proportion of catalogues at quality grade A or above; API uptime (target 99.5%); registered users; downloads and API queries; and peer-reviewed publications citing the platform or a versioned catalogue DOI. Reporting against the FAIR sub-principle indicators (Table 11) should also be included, and annual reports published openly. Illustrative two-year targets: the full GeoNet national catalogue (well over one million events) plus 20 or more additional regional, temporary, and historical catalogues accessible through one interface; 200 or more registered users across the research, engineering, and agency communities; 10 or more research outputs citing a versioned catalogue DOI; and API availability approaching the 99.5% operational target. These are indicative planning targets, not commitments, and should be revisited once the service is established.

5.2 Roadmap

Development proceeds in four phases (Figure 3): planning and mobilisation; prototyping and pilot integration; full-scale implementation and public launch; and sustained operation as a national core service with annual review and FAIR compliance reporting.

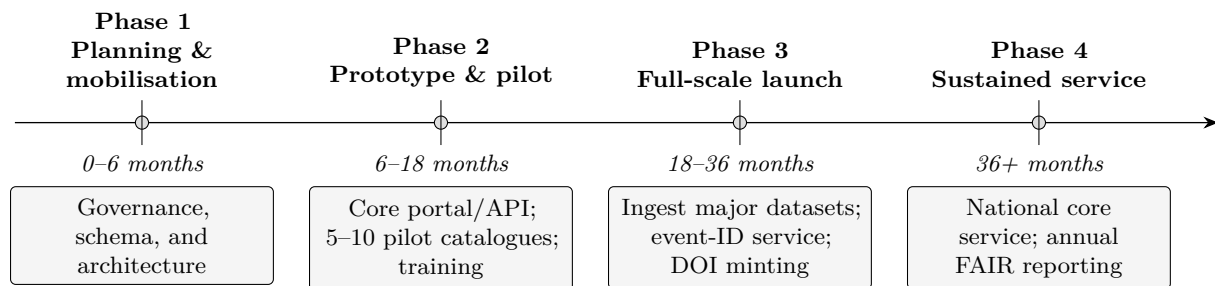


Figure 3: Indicative four-phase roadmap from planning to a sustained national service.

5.3 Funding and Resourcing

Indicative resourcing for a mature service is on the order of three to four full-time-equivalent (FTE) staff (Section 5.1.2) — a Data Stewardship Lead, one to two Data Managers, a Software Engineer, and a part-time User Support Analyst — together with cloud or on-premises infrastructure for storage, compute, and backup, and recurring DOI-registration fees. At New Zealand loaded-salary rates this implies a steady-state operating cost on the order of NZD 0.6–0.9 million per year (lower during the smaller-team Phases 1–2), exclusive of one-off build costs. The natural home is a national research-infrastructure line — for example, an Earth Sciences New Zealand core-function budget or a research-infrastructure programme such as those administered by the Ministry of Business, Innovation and Employment (MBIE) — so that the service is funded as a standing capability rather than a fixed-term project. Indefinite retention of every versioned release is a growing-storage liability; it is bounded by keeping only changelogs and metadata for superseded *patch* releases, retaining full event tables only for DOI-bearing major and minor versions, and moving older releases to tiered/cold storage. The figures here are indicative of scale, not costed estimates; precise budgets should be developed with the host institution.

6 Benefits by Stakeholder

A coordinated, FAIR-and-CARE catalogue capability delivers concrete, audience-specific value:

- **Researchers** gain discoverable, citable, version-pinned datasets and ready cross-catalogue comparison, making studies faster to run and genuinely reproducible.
- **Hazard and risk modellers** (e.g. the National Seismic Hazard Model) get comparable, documented inputs with consistent magnitudes and uncertainties, reducing the bespoke harmonisation that the 2022 NSHM required (Gerstenberger et al., 2024).
- **Engineering and insurance** users obtain trusted, traceable inputs with explicit quality grades and licences, supporting defensible design and risk decisions.
- **Emergency management and the public** benefit from clearer, better-documented event information and a single authoritative point of discovery.
- **Iwi and hapū** are engaged as partners in governance, setting the access and attribution conditions for data of significance to them, which the platform then implements (Section 4.9).
- **The national system** preserves at-risk legacy and temporary datasets and aligns New Zealand with international best practice (EPOS, 2025b).
- **Community infrastructure** gains a stepping-stone, in the shared Registry, toward broader community goals identified at the workshops — digitising and back-filling legacy catalogues, and converging on a national community velocity model (1D and 3D). The Registry is revised on a regular cadence (about every five years) to stay current.

7 Risks and Anticipated Questions

We anticipate, and address, the main objections to this proposal:

- **“Isn’t this duplicating GeoNet?”** No. The framework is an aggregation and curation layer over GeoNet and the many catalogues outside it; GeoNet remains the authoritative operational source and a primary data provider (Figure 2).
- **“Who pays once the establishment funding ends?”** The service is designed as a standing national capability hosted on a core-function or research-infrastructure budget — not a project that ends with a grant. The indicative steady-state cost and a bounded-retention storage policy are given in Section 5 (Funding).
- **“What if researchers don’t contribute their catalogues?”** The contribution process is low-friction and offers tangible incentives — DOIs, citations, increased visibility, and quality feedback — and the system delivers value from GeoNet and pilot catalogues from day one.
- **“Why not just use EPOS, ISC, or ComCat?”** Those systems are complementary and the framework is designed to interoperate with them, but none curates New Zealand’s regional, temporary, and historical catalogues or addresses CARE obligations; a national capability is still required.
- **“Will standardisation flatten legitimate scientific differences?”** No. Alternative solutions are preserved with lineage, regional catalogues are cross-referenced rather than silently merged, and merge strategies are explicit and reproducible.

8 Conclusion and Call to Action

New Zealand has the data, the expertise, and a clear community mandate to make its earthquake catalogues FAIR; what is missing is a coordinating framework and the governance to sustain it. This white paper has set out that framework, benchmarked it against international practice, and demonstrated its feasibility through a working prototype. In doing so it has characterised the catalogue landscape and benchmarked it internationally (aim i), proposed minimum attributes, a metadata profile, and a shared lexicon (aim ii), and set out a gap analysis with near-term community steps (aim iii). The remaining step is collective commitment.

We invite GeoNet and Earth Sciences New Zealand, universities, government agencies, iwi and hapū partners, and the engineering and insurance sectors to:

1. endorse the metadata, lexicon, versioning, and quality standards proposed here as a shared national baseline;
2. commit to establishing the Steering Committee and Data Stewardship Team (Sections 5.1.1 and 5.1.2), with representation that operationalises both FAIR and CARE; and
3. support a multi-year funding case to move from prototype to a sustained national service.

We welcome feedback on this draft and expressions of interest in participating in the governance and catalogue-contribution processes described above. Comments may be directed to the corresponding author, Kenny Graham (Earth Sciences New Zealand).

Data and Code Availability

In keeping with the FAIR principles it advocates, the material supporting this white paper is intended to be openly available. The prototype portal source code, the canonical metadata schema, and the controlled parameter lexicon are released under the MIT licence at <https://github.com/KennyGraham1/catalogofcatalogs> (documentation at <https://catalogofcatalogs.readthedocs.io/>), with tagged releases archived under a Zenodo DOI. A citable version of this white paper will likewise be deposited with a DOI. The earthquake catalogues used as examples are available from their respective providers: the GeoNet national catalogue (GeoNet, 2025a), USGS ComCat (USGS, 2025), and the Swiss Seismological Service (Diehl et al., 2025).

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Funding. [Funding sources and grant numbers to be inserted.]

Author contributions. [Author contribution statement to be inserted.]

A International Benchmark Detail

This appendix expands the summary in Section 2 and Table 2.

A.1 New Zealand (GeoNet)

GeoNet maintains New Zealand’s earthquake catalogue as part of the National Earthquake Information Database, the central repository for NZ seismic events (GNS Science, 2020). The catalogue contains origin time, location, magnitude, and seismic phase data, with each event assigned a unique Public ID (GeoNet, 2025a), spanning instrumentally recorded earthquakes (1930s–present) and historical events from oral/written records (GNS Science, 2020). Data access includes Quake Search (CSV, GeoJSON, KML, GML) (GeoNet, 2025a) and services compliant with the International Federation of Digital Seismograph Networks (FDSN) and Web Feature Service (WFS) standards, delivering QuakeML with uncertainties, phase arrivals, and multiple magnitude determinations (GeoNet, 2025c,b). Completeness is approximately M_L 1–2 in well-monitored regions (higher elsewhere and through time); location quality varies markedly with station coverage, as quantified by Warren-Smith et al. (2025). A 2022 revision moved the catalogue toward more consistent magnitude reporting nationwide (GNS Science, 2020).

A.2 United States (USGS ComCat)

The United States Geological Survey (USGS) operates the federated Advanced National Seismic System (ANSS) Comprehensive Earthquake Catalog (ComCat), covering global earthquakes ($M_{4.5+}$ complete globally; $M_{1-2.5}$ in the US) (USGS, 2025), accessed via the FDSN Event API with QuakeML, CSV/GeoJSON/KML, and real-time GeoJSON feeds (U.S. Geological Survey, 2025). Entries include quality metrics (stations used, azimuthal gap, RMS residuals) and magnitude uncertainty; ComCat integrates 100+ networks, stores multiple solutions per event with one preferred, and hosts products such as ShakeMap and crowdsourced intensity data (USGS, 2025).

A.3 Switzerland (SED)

The Swiss Seismological Service (SED) maintains a high-quality catalogue dating to 1879 (Diehl et al., 2025), with historical events to 250 AD. A dense network achieves M_c 1.0–1.5 completeness; SED introduced updated magnitude scales (M_{Lhc} , 2020), employs 3D velocity models and relative relocation, and provides formal uncertainties via QuakeML since 2013 (Diehl et al., 2025). Data are accessible through FDSN web services, and cross-border collaboration integrates 60+ stations from France, Germany, Italy, and Austria (Diehl et al., 2025). The ECOS project unified historical records into a homogenised database (Fäh et al., 2011).

A.4 European and Global Aggregators

The European-Mediterranean Seismological Centre (EMSC) operates the European Seismic Portal, aggregating data from multiple networks via FDSN-compliant services and providing multiple solutions and phase data (European-Mediterranean Seismological Centre, 2025a,b). The International Seismological Centre (ISC) archives global bulletin data from 130+ agencies, accepting multiple formats (Nordic, GSE, ISF, QuakeML) (International Seismological Centre, 2025a) with output in ISF or QuakeML (International Seismological Centre, 2025b). Other agencies (the Japan Meteorological Agency, Geoscience Australia, and the European Integrated Data Archive) have adopted FDSN services, and major catalogues now share common standards for open, interoperable access (EPOS, 2025b; ETH Zürich Seismology / QuakeML Working Group, 2015).

B Draft Technical Specification

This appendix collects implementation-level detail referenced from Section 4. It is indicative and intended to be refined during Phase 1.

B.1 Metadata Field Hierarchy

Table 6 gives the full catalogue- and event-level metadata profile summarised in Section 4, distinguishing required from recommended fields.

Table 6: Metadata Field Hierarchy for FAIR Catalogue Records

Field	Status	Description
<i>Catalogue-level</i>		
Persistent identifier	Required	DOI or equivalent resolvable identifier
Name, description	Required	Human-readable title and summary
Institution, contact	Required	Responsible party and contact details
Temporal coverage	Required	Start and end date of events
Geographic coverage	Required	Bounding box or region descriptor
Magnitude range	Required	Minimum and maximum magnitude
Event count	Required	Total number of events
Licence	Required	Machine-readable licence identifier (e.g. CC-BY 4.0)
Processing summary	Required	High-level description of methods
Velocity model	Recommended	Reference to velocity model used
Location algorithm	Recommended	Software and version (e.g. NonLinLoc 7.0)
Magnitude calibration	Recommended	Scale definitions and calibration reference
Publication links	Recommended	DOIs of associated papers
<i>Event-level</i>		
Origin time	Required	UTC time (ISO 8601)
Latitude, longitude	Required	WGS84 decimal degrees
Depth	Required	Kilometres below sea level
Magnitude value & type	Required	Value and scale code (e.g. M_w , M_L)
Origin time uncertainty	Recommended	Seconds (1σ)
Horizontal uncertainty	Recommended	Kilometres (1σ)
Vertical uncertainty	Recommended	Kilometres (1σ)
Magnitude uncertainty	Recommended	Standard deviation
Station count	Recommended	Number of stations used
Azimuthal gap	Recommended	Degrees
Phase count	Recommended	Number of phase picks used
RMS residual	Recommended	Seconds
Evaluation status	Recommended	“automatic” or “reviewed”
Event type	Recommended	QuakeML event type vocabulary

B.2 Controlled Parameter Lexicon (Illustrative)

Table 7 shows representative entries from the controlled lexicon referenced in Section 4. It is an illustrative subset — aligned with the QuakeML enumerations (ETH Zürich Seismology / QuakeML Working Group, 2015; Mulder, 2016) — not the complete normative list, which is maintained with the released schema (see Data and Code Availability).

Table 7: Illustrative subset of the controlled parameter lexicon (representative terms; aligned with QuakeML).

Category	Representative controlled terms
Magnitude type	Mw (moment); ML, MLv (local); mb (body-wave); Ms (surface-wave); Md (duration)
Event type	earthquake; induced or triggered event; quarry blast; explosion; volcanic eruption; other event
Phase name	P, S; Pn, Pg, Sn, Sg (crustal); pP, sP (depth phases)
Evaluation mode / status	mode: manual / automatic; status: preliminary / confirmed / reviewed / final / rejected
Pick onset / polarity	onset: emergent / impulsive / questionable; polarity: positive / negative / undecidable
Depth determination	from location; operator assigned; constrained by depth phases; constrained by direct phases
Units & formats	time: ISO 8601 UTC; coordinates: WGS84 decimal degrees; depth: km below sea level

B.3 Versioning Scheme

Table 8 illustrates the version-increment triggers for the three-level MAJOR.MINOR.PATCH scheme described in Section 4.

Table 8: Versioning Scheme: Examples of Version Increment Triggers

Version	Trigger	Example change
MAJOR	Event parameters changed	Reprocessing with new 3D velocity model; recalibration of magnitude scale; new location algorithm affecting systematic offsets
MINOR	New events or fields added	Ingestion of 500 additional aftershock events; addition of focal mechanism fields to existing events
PATCH	Metadata or documentation	Contact email updated; description clarified; deprecated field removed from catalogue-level record

B.4 API Design

A unified API follows REST conventions. Authentication uses token-based credentials for write operations; read operations on public catalogues are unauthenticated. Rate limiting protects service availability, and standard HTTP error codes with a consistent JSON error envelope

enable reliable programmatic error handling. Output format is negotiated via the **Accept** header or a **format** query parameter, defaulting to GeoJSON for spatial queries and CSV for tabular exports; QuakeML output is generated from the canonical internal event model, preserving uncertainty fields, evaluation status, and magnitude type. Table 9 summarises the principal endpoint categories.

Table 9: Principal API Endpoint Categories

Category	Operations	Notes
Catalogues	List, retrieve, create, update, delete	Metadata including bounds, event count, licence
Events	List, spatial query, time-range, magnitude/depth filter	Pagination; supports GeoJSON, CSV, QuakeML output
Search	Cross-catalogue full-text and parameter search	Returns ranked results across all accessible catalogues
Import	Upload file, sync from FDSN source, view history	Supports CSV, JSON, GeoJSON, QuakeML 1.2
Merge	Preview, execute, retrieve conflict log	Dry-run mode returns statistics without committing
Export	Per-catalogue or per-query download	CSV, GeoJSON, KML, QuakeML; version-specific
Versions	List versions, retrieve specific version	Version-specific event snapshots remain available indefinitely

B.5 Access Control and Audit

Access to catalogue data and portal operations is governed by a role-based model. Four *technical* roles are defined (distinct from the governance roles of Section 5.1.3; a single individual may hold both):

- **Administrator:** full access including user management, system configuration, and deletion of catalogue records.
- **Editor:** can create and modify catalogues, import data, and initiate merge operations, but cannot manage users or alter system configuration.
- **Viewer:** can search, view, filter, and export catalogues to which they have been granted access, but cannot modify data.
- **Guest** (unauthenticated): can discover and download public catalogues; embargoed or restricted catalogues are not visible.

Role assignments are managed by Administrators and recorded in an audit log; a role-upgrade request workflow allows registered users to request elevated access.

Data-level access control. Each catalogue carries one of three access designations: **public** (all, unauthenticated), **registered** (authenticated Viewers and above), or **restricted** (named individuals or groups). This supports time-limited embargoes (e.g. allowing a research group to work privately until a companion paper is published). Field-level masking may be applied where a catalogue contains sensitive location information, such as event locations near culturally significant sites: coordinates may be spatially generalised or withheld for Guest and Viewer access while full precision is retained for authorised users, consistent with the CARE discussion in Section 4.9.

Audit and compliance. All write operations (catalogue creation, import, merge, deletion) are recorded with the authenticated user identity, timestamp, and operation details; read access to restricted catalogues is similarly logged. Where usage agreements are attached to a catalogue (for example, a requirement to cite a companion publication), the agreement is presented at download time and acceptance is recorded.

B.6 Uncertainty Representation

Origin location and time uncertainties are reported as standard deviations: `time_uncertainty` (seconds), `horizontal_uncertainty` (kilometres), and `depth_uncertainty` (kilometres). Where horizontal uncertainty is anisotropic, the semi-major and semi-minor axes of the confidence ellipse, its azimuth, and the stated confidence level (conventionally 68%) are reported, mapping to QuakeML `OriginUncertainty` elements (ETH Zürich Seismology / QuakeML Working Group, 2015). Magnitude uncertainties are reported as the standard deviation on the stated scale, and the magnitude type code must always accompany the value. Events without quantified uncertainties carry a boolean `uncertainty_available` field set to false; historical events may substitute a qualitative confidence class. When events are merged, the averaging strategy combines location, time, and magnitude by inverse-variance weighting; because agency solutions for one event are typically correlated rather than independent, the merged uncertainty is reported conservatively — not below the smallest contributing value — rather than the smaller independent-combination result. Priority and quality strategies retain the selected solution’s uncertainty and record alternatives; the merge log documents the strategy and resulting uncertainty.

B.7 Quality Checks and Full Grade Scheme

Representative automated checks include:

- events ordered chronologically, with no future-dated events;
- depth within a physically plausible range for the region — for New Zealand, about -5 to 700 km, covering Hikurangi–Kermadec subduction depths and allowing for events above the sea-level datum (a higher default applies to global catalogues);
- magnitude within a plausible range for the declared region;
- uncertainty values non-negative and within parameter-specific sanity bounds (for example, horizontal uncertainty not exceeding the catalogue’s declared spatial extent, and magnitude uncertainty below a configurable ceiling such as 1.0 unit);
- no coordinate pair repeated for distinct events at an identical time (duplicate detection);
- station count ≥ 5 for an event to be considered well-constrained; and
- azimuthal gap $< 180^\circ$ for a reliably constrained epicentral location (USGS, 2025).

Results are summarised as passed, warning, or failed. The full eight-band grade scheme underlying the summary in Table 5 is given in Table 10.

Table 10: Full Quality Grade Thresholds and Recommended Use (illustrative; subject to community calibration)

Grade	Score	Interpretation and recommended use
A+	95–100	Excellent. Publication-ready; suitable for hazard modelling and all quantitative analysis.
A	85–94	Good. Suitable for most research and engineering applications.
B+	75–84	Acceptable. Minor gaps; document limitations in methods sections.
B	65–74	Fair. Notable gaps; consider supplementing with additional metadata.
C+	55–64	Poor. Significant gaps; expert review recommended before quantitative use.
C	45–54	Very poor. Limited utility for quantitative analysis.
D	35–44	Minimal utility. Use only for discovery or illustrative purposes.
F	<35	Not recommended for quantitative analysis.

B.8 Special Catalogue Type Handling

Regional research catalogues produced with specialised methods (e.g. 3D velocity models, relative relocation, custom magnitude calibrations) are ingested as distinct catalogues with their own version records and DOIs; processing methods are documented in catalogue-level metadata, and the portal provides cross-reference links between co-temporal events in regional and national catalogues with the spatial and magnitude offset reported. **Temporary deployment catalogues** are ingested with an explicit time window and a “research” or “monitoring” designation, with station-geometry metadata and links to relevant FDSN waveform services. **Historical and pre-instrumental catalogues** are accommodated through qualitative location confidence classes, macroseismic intensity fields (Modified Mercalli or EMS-98, Grünthal, 1998), narrative source-attribution fields, and separate quality thresholds, and are presented as a separate collection. **Derived and synthetic catalogues** carry mandatory provenance identifying source catalogue(s), derivation method, parameters, and software, with explicit “derivedFrom” relations to source versions; synthetic catalogues are labelled as such.

B.9 FAIR Compliance Matrix

Table 11 gives the full mapping summarised in Section 4: each FAIR sub-principle, the framework mechanism that addresses it, and a measurable indicator for annual reporting.

Table 11: Illustrative FAIR Compliance Matrix: Sub-Principle, Framework Mechanism, and Indicator

Sub-pr.	Framework mechanism	Indicator
F1	Persistent identifiers (DOIs) for catalogues; stable event identifiers	% of catalogues with a resolvable DOI; % of events with a stable ID
F2	Rich metadata profile (catalogue- and event-level)	Mean metadata-completeness score across catalogues
F3	Metadata records embed the data identifier	% of metadata records linking to their data identifier
F4	Indexing in the searchable portal and discovery API	% of catalogues indexed and discoverable
A1	Retrieval via FDSN/REST over HTTPS	API uptime; % of catalogues retrievable by identifier
A1.1/A1.2	Open, standard protocols; token auth only for write/restricted data	% of read endpoints requiring no authentication
A2	Metadata and version records retained after data withdrawal	% of withdrawn datasets with surviving metadata
I1	QuakeML / FDSN standard representations	% of catalogues available as valid QuakeML
I2	Controlled parameter lexicon (magnitude, event, phase types)	% of fields using controlled-vocabulary terms
I3	Qualified cross-references (event-ID mapping, lineage links)	% of merged events with recorded source lineage
R1.1	Machine-readable licence on every record	% of catalogues with a machine-readable licence
R1.2	W3C PROV-based provenance (entities, activities, agents)	Mean provenance-completeness score
R1.3	Conformance to FDSN/QuakeML community standards	% of catalogues passing standards-compliance checks

C Summary of Workshop Themes

This appendix summarises the workshop discussions, synthesised here as nine cross-cutting themes. The content reflects community perspectives on minimum catalogue attributes, metadata requirements, processing concerns, governance needs, and priorities for a national FAIR-aligned framework for earthquake catalogues in Aotearoa New Zealand.

Table 12: Summary of Key Themes from the Workshop Discussions

Theme	Key Points
Minimum Catalogue Attributes	Phase picks; station metadata; velocity models (1D/3D); depths and magnitudes with uncertainties; completeness estimates; region of interest; unique IDs; focal mechanisms; processing history; detection and processing methods.
Metadata & Documentation Needs	Clear methodology (location algorithm, velocity model, filters); solution provenance; waveform availability; descriptive summaries; parameter definitions; assumptions (point-source vs 2D/3D); embargo details; QC information; required README and DOI; multiple contacts.
Access & Versioning	Raw data accessibility; open formats (CSV, QuakeML, FDSN); programmatic access (API); search by region, time, magnitude; support for multiple versions; backward/forward compatible formats; tracking of method changes; long-term storage of legacy releases.
Governance Needs	Defined roles (contributor, reviewer, maintainer); central catalogue portal; consistent naming conventions; periodic updates; upload validation; framework transparency; responsibility for long-term stewardship.
Community Priorities	National catalogue portal; common parameter lexicon; national 3D community velocity model; honest uncertainties; digitisation & reprocessing of legacy data; cross-catalogue interoperability; clear documentation for older datasets.
Concerns Raised	Inconsistent completeness; variable processing; missing metadata; uncertain magnitude/depth scales; “black-box” ML outputs; compatibility issues across regional/temporary catalogues; handling of unpublished datasets; need for quality flags and user warnings.
Desired Portal Features	Merge multiple catalogues; support subcatalogs; synthetic catalogues; geographic reference tools; method-based filtering; visualisation; minimum-requirements upload form; linkage among event, waveform, and station data.
What Users Liked	Simple form; unified access; clear uncertainty fields; reproducibility focus; increased awareness of catalogue availability; prioritisation of essential fields.
Requested Improvements	More metadata fields and filters; better uncertainty representation; link with 2D/3D models; clarify naming/versioning; transparent ML integration; improved handling of old locations; inclusion of focal mechanisms and strong-motion connections.

D Glossary

ANSS / ComCat Advanced National Seismic System; its Comprehensive Earthquake Catalog (USGS).

Azimuthal gap the largest angle, seen from an epicentre, between adjacent recording stations; a smaller gap means a better-constrained location.

CARE Collective benefit, Authority to control, Responsibility, Ethics — principles for Indigenous data governance.

CMT / GCMT centroid moment tensor; GCMT is the Global CMT project’s moment-tensor catalogue.

Conflation / merge combining records of the same earthquake from different catalogues into one unified event.

Declustering separating mainshock–aftershock sequences from background seismicity (e.g. the Gardner–Knopoff method).

DOI Digital Object Identifier — a persistent, citable identifier for a dataset or document.

EMSC / ISC European-Mediterranean Seismological Centre; International Seismological Centre.

EPOS European Plate Observing System — a coordinated FAIR data infrastructure for solid-Earth science.

FAIR Findable, Accessible, Interoperable, Reusable — principles for scientific data management.

FDSN International Federation of Digital Seismograph Networks; its web services are a global standard for seismic data access.

GeoNet New Zealand’s geological-hazard monitoring programme, operated by Earth Sciences New Zealand.

M_c magnitude of completeness — the lowest magnitude above which all earthquakes in a catalogue are reliably recorded.

M_w , M_L moment magnitude and local (Richter) magnitude — two magnitude scales.

NSHM (Aotearoa New Zealand) National Seismic Hazard Model.

OBS ocean-bottom seismometer — a seafloor instrument used in temporary offshore deployments.

PROV / PROV-O the W3C provenance data model and ontology (entities, activities, and agents).

Provenance the documented origin and processing history of a dataset (inputs, methods, software, agents).

PSHA probabilistic seismic hazard analysis.

QuakeML / StationXML standard XML formats for exchanging earthquake parameters and station metadata.

WFS Web Feature Service — an OGC standard for serving geographic features.

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